ELTE IK - Computer Science BSc

Final Exam Topics

1. Limits and continuity of functions

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Limits and continuity of functions

Limits of functions, continuity. Properties of functions that are continuous on a compact set: Weierstrass theorem, Bolzano theorem. The concept of a power series, Cauchy–Hadamard theorem, analytic functions.

1 Limits of functions

Given $f \in \mathbb{K}_1 \to \mathbb{K}_2, a \in \mathcal{D}'_f$ (accumulation point). The function f has a limit at the point a if $\exists A \in \overline{\mathbb{K}}_2$, where $\overline{\mathbb{K}} = \mathbb{C} \vee \mathbb{R} \vee +\infty \vee -\infty$, such that for any neighborhood $K(A) \subset \mathbb{K}_2$ a suitable neighborhood $k(a) \subset \mathbb{K}_1$ of a can be given with which $f[(k(a)\backslash\{a\}) \cap \mathcal{D}_f] \subset K(A)$ holds.
Otherwise: $f(x) \in K(A), (a \neq x \in k(a) \cap \mathcal{D}_f)$.
Then the uniquely existing number $A \in \overline{\mathbb{K}}_2$ or one of the infinities is called the limit of the function f at the point a.
Notation: $\lim\limits_{x \to a} f(x) = \lim\limits_{a} f = A$

1.1 Accumulation point

$A \subset \mathbb{K}$, then $\alpha \in \overline{\mathbb{K}}$ is an accumulation point of the set A if for any neighborhood $K(\alpha)$ we have $(K(\alpha)\backslash\{\alpha\}) \cap A \neq \emptyset$.
With inequalities: $A \subset \mathbb{K}$, then the number $\alpha \in \mathbb{K}$ is an accumulation point of the set A if for $\forall \varepsilon > 0$ there exists $x \in A$ such that $0 < |x - \alpha| < \varepsilon$.

1.2 Neighborhood

$A \subset \mathbb{K}, a \in A, r > 0 : K_r(a) = K(a) = \{x \in A : |x - a| < r\}$.

1.3 Function

$\emptyset \neq A, B$ sets, $f \subset A \times B$ a relation. For some element $x \in A$ let $f_x := \{y \in B : (x, y) \in f\}$. The relation f is a function if for $\forall x \in A$ the set f_x has at most one element.

2 Continuity of functions

The function $f \in \mathbb{K}_1 \to \mathbb{K}_2$ is continuous at the point $a \in \mathcal{D}_f$ if for $\forall \varepsilon > 0$ a number $\delta > 0$ can be given with which for any $x \in \mathcal{D}_f, |x - a| < \delta$ we have $|f(x) - f(a)| < \varepsilon$.
Notation: $f \in \mathcal{C}\{a\}$; if $\forall x \in \mathcal{D}_f : f \in \mathcal{C}\{x\}$, then $f \in \mathcal{C}$.

3 Relationship between limit and continuity

Let $f \in \mathbb{K}_1 \to \mathbb{K}_2, a \in \mathcal{D}_f \cap \mathcal{D}'_f$. Then $f \in \mathcal{C}\{a\} \iff \lim\limits_a f = f(a)$.

4 Concepts

4.1 Compact set

$A \subset \mathbb{K}$ is compact if any sequence $(x_n) : \mathbb{N} \to A$ has a subsequence (x_{ν_n}) that is convergent and $\lim (x_{\nu_n}) \in A$. Then A is bounded and closed.

4.2 Convergent

A number sequence $x = (x_n) : \mathbb{N} \to \mathbb{K}$ is called convergent if $\exists \alpha \in \mathbb{K}, \forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \in \mathbb{N}, n > N : |x_n - \alpha| < \varepsilon$. α is the limit of the sequence x.

4.3 Bounded

For a sequence: x_n bounded $\Rightarrow \exists \nu : x \circ \nu$ convergent
For a set: $\emptyset \neq A \subset \mathbb{K}$ is bounded if $\exists q \in \mathbb{R} : |x| \leq q, (x \in A)$

4.4 Closed set

Its complement is an open set.

4.5 Open set

A is an open set $\iff \forall a \in A, \exists K(a) : K(a) \subset A$, that is, every point of the set A is an interior point.

5 Heine theorem (optional)

If the function $f \in \mathbb{K}_1 \to \mathbb{K}_2$ is continuous and $\mathcal{D}_f$ is compact, then f is uniformly continuous.

5.1 Uniformly continuous

The function $f \in \mathbb{K}_1 \to \mathbb{K}_2$ is uniformly continuous if $\forall \varepsilon > 0, \exists \delta > 0 : |f(x) - f(t)| < \varepsilon, (x, t \in \mathcal{D}_f, |x - t| < \delta)$.

6 Weierstrass theorem

Suppose that the function $f \in \mathbb{K} \to \mathbb{R}$ is continuous and $\mathcal{D}_f$ is compact. Then the range $\mathcal{R}_f$ has a greatest and a smallest element ($\exists \max \mathcal{R}_f, \exists \min \mathcal{R}_f$).

7 Bolzano theorem

Suppose that for some $-\infty < a < b < +\infty$ the function $f : [a, b] \to \mathbb{R}$ is continuous, and $f(a) \cdot f(b) < 0$ (of opposite sign).
Then there exists $\xi \in (a, b)$ such that $f(\xi) = 0$.

8 The concept of a power series

Let the center $a \in \mathbb{K}$ and the coefficient sequence $(a_n) : \mathbb{N} \to \mathbb{K}$ be given, and using these consider the following functions: $f_n(t) := a_n(t - a)^n, (t \in \mathcal{D} := \mathbb{K}, n \in \mathbb{N})$. Then the function series $\sum (f_n)$ is called a power series.

8.1 Sequence

The function f is called a sequence if $\mathcal{D}_f = \mathbb{N}$.

8.2 Function sequence, function series

A sequence (f_n) is a function sequence if for $\forall n \in \mathbb{N}$ each f_n is a function, and for some set $\mathcal{D} \neq \emptyset$ we have $\mathcal{D}_{f_n} = \mathcal{D}, (n \in \mathbb{N})$.
If for the function sequence (f_n) in question $\mathcal{R}_{f_n} \subset \mathbb{K}, (n \in \mathbb{N})$ also holds, then the function series $\sum(f_n)$ determined by the function sequence (f_n) is the following function sequence: $\sum(f_n) := (\sum\limits_{k=0}^{n} f_k)$.

9 Cauchy–Hadamard theorem

Suppose that for the sequence $(a_n) : \mathbb{N} \to \mathbb{K}$ the limit $\lim(\sqrt[n]{|a_n|})$ exists, and let

$$r := \begin{cases} +\infty & \text{if } \lim(\sqrt[n]{|a_n|}) = 0 \\ \dfrac{1}{\lim(\sqrt[n]{|a_n|})} & \text{if } \lim(\sqrt[n]{|a_n|}) > 0 \end{cases}$$

We call r the radius of convergence. Then for any $a \in \mathbb{K}$ we can say the following about the power series $\sum(a_n(t-a)^n)$:

1. If $r > 0$, then at every point $x \in \mathbb{K}, |x-a| < r$ the power series $\sum(a_n(t-a)^n)$ is absolutely convergent at x.
2. If $r < +\infty$, then for any $x \in \mathbb{K}, |x-a| > r$ the power series $\sum(a_n(t-a)^n)$ is divergent at x.

9.1 Absolutely convergent

The infinite series $\sum\limits_{n=1}^{\infty} a_n$ is called absolutely convergent if the series $\sum\limits_{n=1}^{\infty} |a_n|$ is convergent.

9.2 Divergent

If $\lim\limits_{n\to\infty} a_n$ does not exist, or is not finite, then the infinite series $\sum a_n$ is divergent.

9.3 Consequences of the Cauchy–Hadamard theorem

1. If $r = +\infty$, then $\mathcal{D}_0 = \mathbb{K}$, and for $\forall x \in \mathbb{K}$ the power series is absolutely convergent.
2. If $r = 0$, then $\mathcal{D}_0 = \{a\}$ and $\sum\limits_{n=0}^{\infty} a_n(a-a)^n = a_0$.
3. If $0 < r < +\infty$, then $K_r(a) \subset \mathcal{D}_0 \subset G_r(a) = \{x \in \mathbb{K} : |x-a| \leq r\}$ and at every point x of the neighborhood $K_r(a)$ the power series is absolutely convergent at x.
4. If $r > 0$, then for any number $0 \leq \rho < r$ choose the element $x \in K_r(a)$ such that $|x-a| = \rho$. Then $\sum\limits_{n=0}^{\infty} |a_n(x-a)^n| = \sum\limits_{n=0}^{\infty} |a_n||x-a|^n) = \sum\limits_{n=0}^{\infty} |a_n|\rho^n) < +\infty$

10 Analytic functions

Suppose that the radius of convergence r (see C–H theorem) of the power series $\sum(a_n(t-a)^n)$ is not zero. Then we can define the following function: $f(x) := \sum\limits_{n=0}^{\infty} a_n(x-a)^n, (x \in K_r(a))$, which is nothing other than the restriction of the sum function $F(x) := \sum\limits_{n=0}^{\infty} a_n(x-a)^n, (x \in \mathcal{D}_0)$ of the function series $\sum(a_n(t-a)^n)$ to the neighborhood $K_r(a)$: $f = F|_{K_r(a)}$, (f is the sum function of the power series). A function f of such a structure is called analytic.
Remark: $\mathcal{D}_0$ denotes the domain of convergence of the power series $\sum(a_n(t-a)^n)$.

10.1 Important analytic functions

Coefficients a_n for which $\lim\left(\sqrt[n]{|a_n|}\right) = 0$, hence $r = +\infty$ for the power series $\sum(a_n t^n)$.

These a_n are: $\dfrac{1}{n!}, \dfrac{(-1)^n}{(2n+1)!}, \dfrac{(-1)^n}{(2n)!}, \dfrac{1}{(2n+1)!}, \dfrac{1}{(2n)!}$.

- $\exp x := \exp(x) = e^x = \sum\limits_{n=0}^{\infty} \dfrac{x^n}{n!}, (x \in \mathbb{K})$
- $\sin x := \sin(x) = \sum\limits_{n=0}^{\infty} (-1)^n \dfrac{x^{2n+1}}{(2n+1)!}, (x \in \mathbb{K})$
- $\cos x := \cos(x) = \sum\limits_{n=0}^{\infty} (-1)^n \dfrac{x^{2n}}{(2n)!}, (x \in \mathbb{K})$
- $\sinh x := \sinh(x) = \sum\limits_{n=0}^{\infty} \dfrac{x^{2n+1}}{(2n+1)!}, (x \in \mathbb{K})$
- $\cosh x := \cosh(x) = \sum\limits_{n=0}^{\infty} \dfrac{x^{2n}}{(2n)!}, (x \in \mathbb{K})$